

ASHISH PRATAP SINGH

Full Stack Web Developer | MERN Stack & TypeScript

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TECHNICAL HIGHLIGHTS

- **Full-Stack MERN & TypeScript Engineer** with 3 production-grade applications live on Vercel.
- **Architectural Focus:** Low-latency AI integration (OpenRouter), geospatial routing (OSRM/Leaflet), and custom audio state management.
- **Production Standards:** Secure RESTful API design (JWT/bcrypt/CORS), MongoDB indexing, and sub-1.2s Core Web Vitals performance.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL

Frontend: React.js (v18/19), Vite, Tailwind CSS (v4), Responsive UI, Context API, PWA

Backend: Node.js, Express.js, RESTful API Design, JWT Auth, Middleware Architecture, bcrypt

Databases & Cloud: MongoDB (Mongoose, Aggregations, Indexing), PostgreSQL, Supabase, Vercel CI/CD

APIs & Tooling: OpenRouter AI, YouTube IFrame API, Leaflet / OSRM, Postman API Testing, Git/GitHub

TECHNICAL PROJECTS

InterviewIQ — AI Mock Interview & Career Prep Platform

[\[Live Demo\]](#) | [\[GitHub\]](#)

React 19, Tailwind CSS v4, Node.js, Express, MongoDB, OpenRouter AI, Vercel

- **Architected sub-250ms voice-to-text pipeline** by coupling Web Audio streaming listeners with OpenRouter AI APIs, enabling natural, low-latency mock interview dialogue.
- **Engineered automated evaluation engine** generating structured 5-category performance cards and downloadable PDFs within 3 seconds of interview completion.
- **Secured session persistence & transcript history** across test users by designing normalized MongoDB schemas with compound indexing and custom Express validation middleware.

SONIQ — AI-Powered Music Streaming & Queue Platform

[\[Live Demo\]](#) | [\[GitHub\]](#)

React, TypeScript, Vite, Tailwind CSS, Supabase (PostgreSQL), YouTube API, PWA

- **Cut audio hosting & CDN bandwidth costs to \$0** by engineering an embedded playback controller driving real-time audio streams via the YouTube IFrame Player API.
- **Engineered dynamic AI DJ feature** that converts natural language mood prompts (e.g., 'focus lo-fi deep work') into instant auto-playing queues in <4 seconds.
- **Eliminated 100% of playback stutter & desync** across multi-page client transitions by centralizing queue state inside a dedicated React Context with strict action dispatchers.
- **Accelerated initial mobile load time by 35%** by implementing offline Service Worker PWA caching strategies and Vite production bundle tree-shaking.

Uber Clone — On-Demand Ride Dispatch & Geospatial Platform

[\[Live Demo\]](#) | [\[GitHub\]](#)

React, Node.js, Express, MongoDB, JWT, Leaflet, OSRM, Tailwind CSS

- **Engineered dual-role access control** with strict boundary isolation between rider and captain portals using role-based JWT auth middleware (authUser, authCaptain).
- **Rendered turn-by-turn route maps in <150ms** by integrating OpenStreetMap (Leaflet) with OSRM routing engines and Nominatim geocoding APIs.
- **Enforced secure 4-step ride state machine** (Requested → Accepted → In-Transit → Completed) using OTP handshake verification, bcrypt password salting, and token blacklisting on logout.

PROFESSIONAL EXPERIENCE

Full-Stack Software Developer (Independent / Production Work)

Aug 2023 – Present

- **Shipped 3 production full-stack web applications** with 99.8% uptime, managing the complete SDLC from Figma UI concepts to automated Vercel CI/CD cloud deployments.
- **Hardened API security & throughput** by implementing Express CORS policies, rate-limiting guards against DoS attacks, and centralized asynchronous error-handling middleware.
- **Optimized MongoDB query performance by ~30%** by writing targeted Mongoose aggregation pipelines and indexing high-frequency lookup fields (IDs, session tokens).
- **Achieved 92+ Google Lighthouse performance scores** across all deployments through asset minification, lazy-loaded route splitting, and Tailwind CSS utility architectures.

EDUCATION

Bachelor of Computer Applications (BCA) — IPS College, Jaipur (Affiliated to Rajasthan University)

2023 – 2026 (Expected)

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Web Engineering, Computer Networks.